DOCUMENTATION

POMODORO TIMER

TRIOE

INTRODUCTION

The POMODORO Timer Kit is an ESP32-based smart time management device created to help students improve their study habits using the Pomodoro Technique. This technique divides work into structured intervals of focused activity followed by short breaks, allowing users to maintain concentration while preventing mental exhaustion.

The project was developed in response to common challenges experienced by students, such as procrastination, difficulty maintaining attention, and inefficient study routines. As digital devices become essential tools for learning, they also introduce distractions that can interrupt focus and reduce productivity.

To address this issue, the POMODORO Timer Kit provides a dedicated physical timer that guides users through structured work sessions and break periods. The device combines simple hardware components—including an ESP32 microcontroller, OLED display, buttons, and a buzzer—with an easy-to-use interface to support effective time management during study sessions.

Importance

The POMODORO Timer Kit serves as a practical tool that helps students structure their study sessions in a consistent and manageable way. By guiding users through timed focus periods and scheduled breaks, the device encourages discipline and promotes a more organized approach to studying.

Benefits

- Improves focus by minimizing distractions
- Prevents mental fatigue by incorporating regular breaks
- Enhances productivity through structured work intervals
- Encourages time awareness when completing tasks
- Helps maintain work-life balance

FEATURES

Key Features

OLED Display Interface

The timer uses a small OLED display to clearly show the countdown time, system status, and timer progress.

WiFi Configuration Setup

Users can connect to the timer using ESP32 WiFi to configure timer settings through a simple web dashboard.

Web-Based Dashboard

The device provides a built-in web interface where users can:

- Enter name
- Set total study duration
- Set study and break duration
- Save the study session

Visual Timer Indicator

A progress bar is displayed on the screen to show the remaining time of the current session.

Buzzer Notification

The timer produces a short beep sound when a session ends to notify the user.

Compact and Portable Design

The device is small and lightweight, making it ideal for desks, study areas, or workspaces.

Type-C Powered

The Pomodoro Timer Kit uses a USB Type-C cable for power, allowing easy connection to power banks, laptops, or USB adapters.

Easy-to-Use Controls

Physical buttons allow users to start the timer or interact with the device quickly.

INSTRUCTIONS

Opening the Pomodoro Kit and Dashboard

1. Plug a **Type-C cable** into the Pomodoro Kit and connect it to a power source
2. Press the **On/Off button** to turn on the Pomodoro Kit.
3. Connect your phone or laptop to the ESP32 WiFi.
   SSID: Pomodoro Timer
   Pass: 12345678
4. Open a browser and type: **192.168.4.1**
5. The Pomodoro Web Dashboard will load.

Dashboard Navigation

1. Enter your name on the first screen.
2. Click **Next.**
3. Select how you are feeling: Motivated, Productive, Tired, or Stressed.
4. If you selected **Tired or Stressed**, read the suggestion page and click Continue to proceed.
5. If you selected **Motivated or Productive**, you will go directly to the timer setup page.
6. Set your:
   - Total Study Time
   - Focus Duration
   - Break Time
7. Click **Save** to confirm your settings.
8. Click **Reset** if you want to clear all input fields.
9. Use the **Back button** anytime to return to the previous page.
10. Press the physical **Start/Pause button** on the Pomodoro Kit to begin.

TROUBLESHOOTING

The device does not turn on

Possible Solutions:

- Check if the USB Type-C cable is properly connected
- Ensure the power source is working
- Try a different USB port or power adapter

I cannot see the WiFi network "Pomodoro Timer"

Possible Solutions:

- Make sure the device is powered on
- Restart the device
- Move closer to the timer

The dashboard page does not open

Possible Solutions:

- Ensure you are connected to the Pomodoro Timer WiFi
- Re-enter the address: 192.168.4.1
- Refresh the browser page

The timer is not starting

Possible Solutions:

- Ensure that the timer settings are saved in the dashboard
- Press the Start Button again

No sound from the buzzer

Possible Solutions:

- Check if the timer session has fully ended
- Restart the device

UPDATING FIRMWARE

The Pomodoro Timer firmware can be updated using the **TRIOE Web IDE**, which allows users to upload new programs to the device through a web interface.

Steps to Update the Firmware

1. Connect the Pomodoro Timer to your computer using the USB Type-C cable.
2. Open the TRIOE Web IDE in your browser.
3. Select the correct device port connected to the timer.
4. Upload or paste the updated firmware .
5. Click Compile/Flash to start the firmware update.
6. Wait until the upload process finishes.
7. Restart the device after the update.

The Pomodoro Timer will now run the new firmware version.